Algebra-0 2026 Spring

Homework 5

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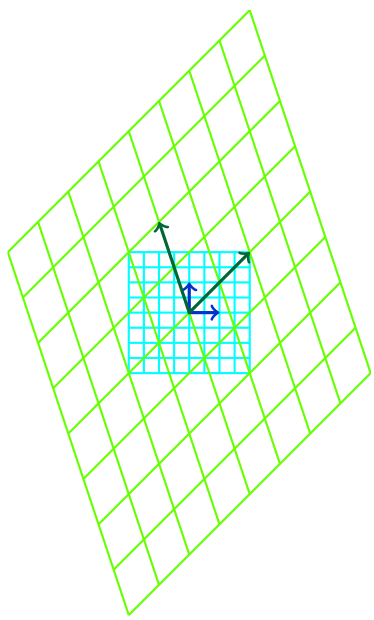
*All of my homework is available at `https://github.com/lacampanella233/Homework.git`.

1 Problems form Artin

Problem 1 (I.2.6). Sketch the effect on the plane $\mathbb{R}^2$ of multiplication by the matrix

$$A = \begin{bmatrix} 2 & -1 \\ 2 & 3 \end{bmatrix}.$$

Solution.



Problem 2 (I.2.10). Let A be a square matrix. Prove that there exist elementary matrices $E_1, \ldots, E_k$

such that

$$E_k \cdots E_1 A$$

is either the identity matrix or has its bottom row equal to zero.

Proof. If A is invertible, then one can transform A to I by elementary row transformation, thus there exists elementary matrices $E_1, \cdots, E_k$ such that $E_k \cdots E_1 A = I$.

If A is not invertible, then by elementary row transfirmation, we can transform A to a row-echelon form, which has its bottom row equal to zero. Thus there exists elementary matrices $E_1, \cdots, E_k$ such that $E_k \cdots E_1 A$ has its bottom row equal to zero. □

Problem 3 (I.4.3). Prove that a matrix in which each row contains exactly one entry equal to 1, each column also contains exactly one entry equal to 1, and all other entries are 0, is a permutation matrix.

Proof. Denote the matrix by A. By the definition of A, we can write

$$A = \sum_{i=1}^{n} E_{i,\sigma(i)}, \ \ \sigma(i) \in \{1, 2, \cdots, n\}.$$

Moreover, since each column of A contains exactly one entry equal to 1, we have $\sigma(i) \neq \sigma(j)$ for $i \neq j$. Thus σ is a permutation of $\{1, 2, \cdots, n\}$, i.e. $\sigma \in S_n$. Therefore A is a permutation matrix. □

Problem 4 (I.4.6). What is the permutation matrix corresponding to the permutation $i \mapsto n - i$?

Solution.

$$\begin{pmatrix} 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 1 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 1 & \cdots & 0 & 0 \\ 1 & 0 & \cdots & 0 & 0 \end{pmatrix}.$$

Problem 5 (I.5.3). Let $A = (a_{ij})$ be an $n \times n$ matrix with integer entries. Prove that A^{-1} has integer entries if and only if

$$\det(A) = \pm 1.$$

Proof. If A^{-1} has integer entries, then $\det A^{-1} \in \mathbb{Z}$. Since $\det(A^{-1}) = (\det A)^{-1}$, we have $(\det A)^{-1} \in \mathbb{Z}$, which implies $\det A = \pm 1$.

Conversely, if $\det A = \pm 1$, then $A^{-1} = (\operatorname{adj} A)/(\det A)$ has integer entries since $\operatorname{adj} A$ has integer entries and $\det A = \pm 1$. □

2 Problems from LADR

Problem 6 (4-10). For $p \in \mathcal{P}(\mathbb{R})$, define $Tp : \mathbb{R} \to \mathbb{R}$ by

$$(Tp)(x) = \begin{cases} \dfrac{p(x) - p(3)}{x - 3}, & \text{if } x \neq 3, \\ p'(3), & \text{if } x = 3. \end{cases}$$

for each $x \in \mathbb{R}$. Show that $Tp \in \mathcal{P}(\mathbb{R})$ for every polynomial $p \in \mathcal{P}(\mathbb{R})$ and also show that $T : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ is a linear map.

Proof. Since the polynomial $p(x) - p(3)$ vanishes at $x = 3$, it has a root $x = 3$, hence a factor $x - 3$. Write

$$p(x) - p(3) = q(x)(x - 3).$$

Take derivative of the equation above and set $x = 3$, we can get $p'(3) = q(3)$. Then $T(p) = q \in \mathcal{P}(\mathbb{R})$.

Let's prove that T is a linear map.

- for $p, q \in \mathcal{P}(\mathbb{R})$,

$$T(p+q) = \frac{p(x) + q(x) - p(3) - q(3)}{x - 3} = \frac{p(x) - p(3)}{x - 3} + \frac{q(x) - q(3)}{x - 3} = T(p) + T(q).$$

- for $p \in \mathcal{P}(\mathbb{R})$ and $c \in \mathbb{R}$,

$$T(cp) = \frac{cp(x) - cp(3)}{x - 3} = c\frac{p(x) - p(3)}{x - 3} = cT(p).$$

Therefore T is a linear map. $\square$